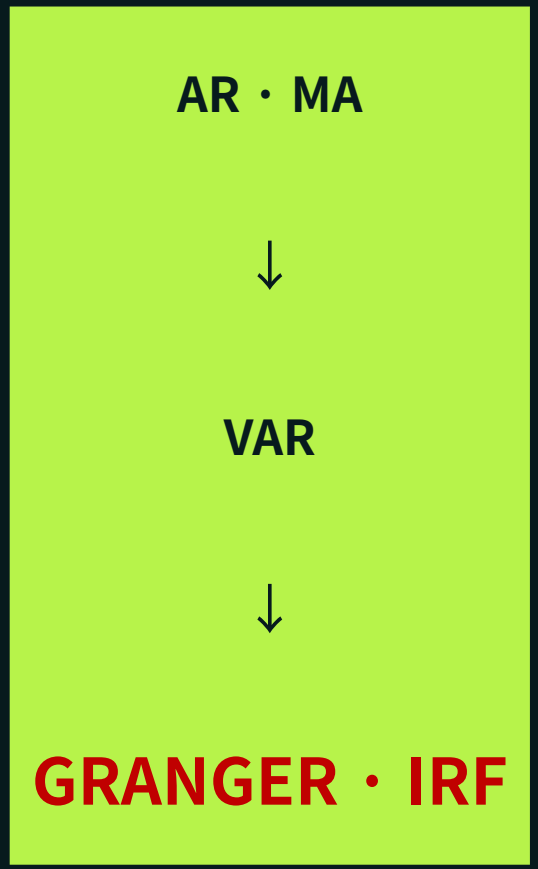
VAR Models: Basics
Several series together

From AR · MA review to VAR · Granger causality · IRF



Advanced Financial Time Series · Week 6

Start from known models and widen step by step

One variable (AR · MA) → many variables (VAR) → two questions for a VAR

00	**Review**	**AR and MA** Forecast from past values, errors
01	**VAR model**	**Many variables at once** Written out · matrices · subscripts · VARMA
02	**Granger causality**	**Does another series' past help?** ADL · $\gamma \neq 0$ · F-test · case · limits
03	**Impulse response · FEVD**	**How a shock spreads** Definition · by hand · Cholesky · order

The only new idea is 'many variables'

00

AR and MA Review

Two ingredients for one variable — past values, past errors

AR forecasts today from its own past values (lags)

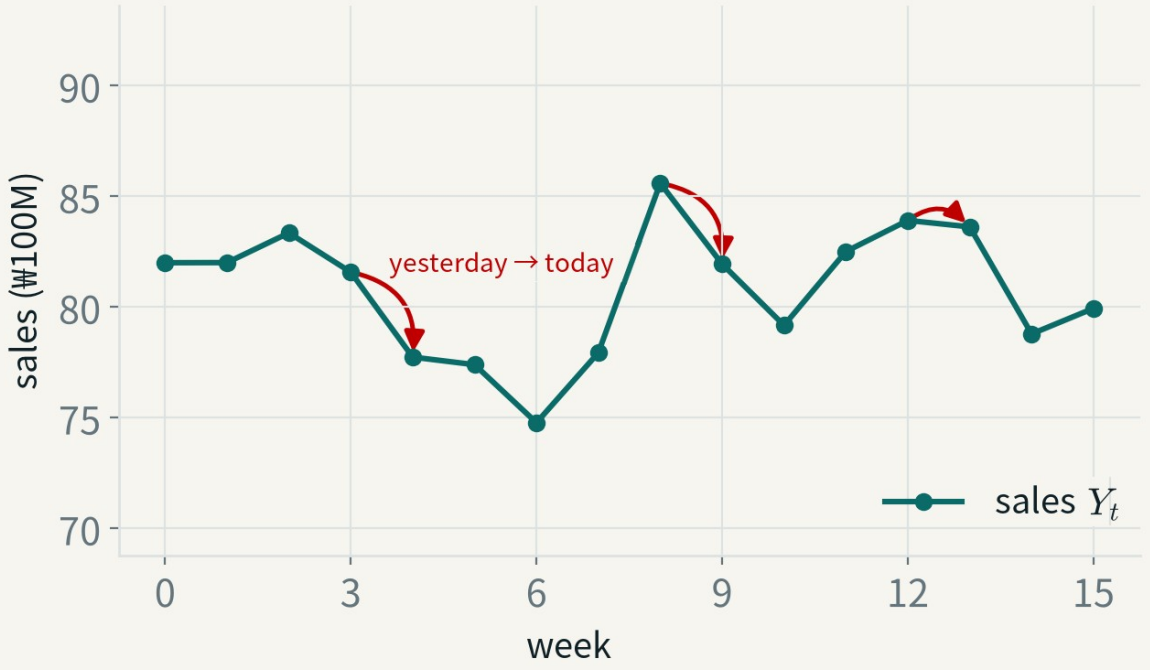
AR = AutoRegressive — past values are called lags

AR(p) model

$$Y_t = \omega + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + e_t$$

How to read it

- Y(t−1): yesterday's value = lag 1
- ϕ: weights; e(t): today's new error



AR's ingredient is 'my own past values'

MA forecasts today from past forecast errors (error lags)

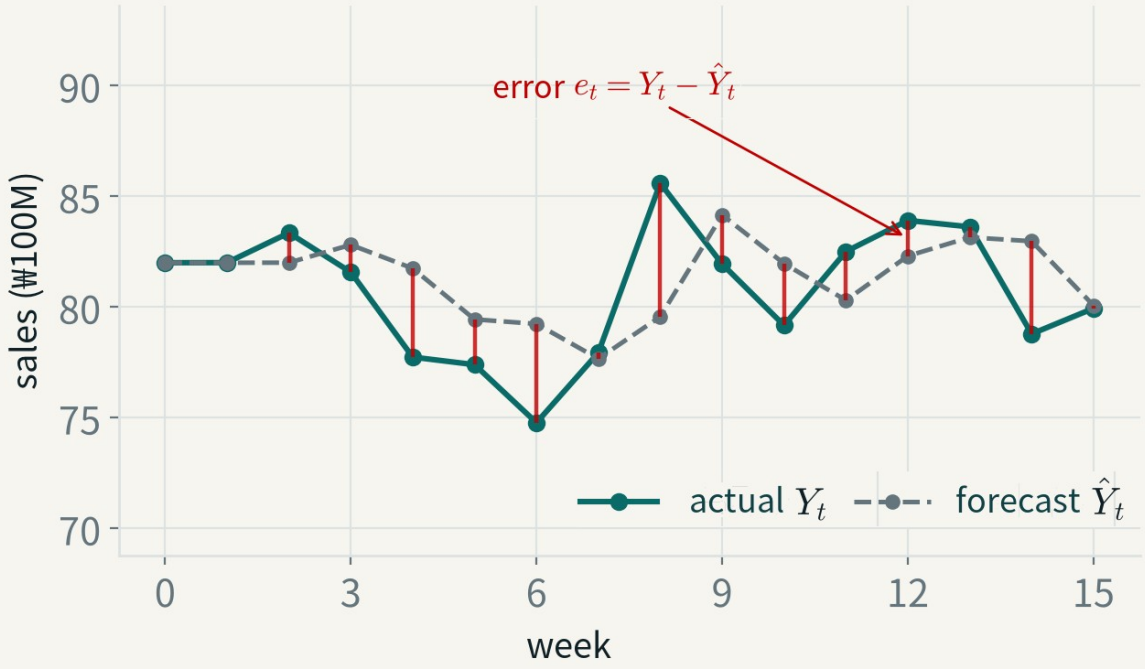
MA = Moving Average — past errors are called error lags

MA(q) model

$$Y_t = \omega + \theta_1 e_{t-1} + \theta_2 e_{t-2} + \cdots + \theta_q e_{t-q} + e_t$$

How to read it

- Error = actual − forecast (red bars)
- Correct today by yesterday's miss



MA's ingredient is 'my own past errors'

ARMA uses past values and past errors together

ARMA(p, q) = AR(p) + MA(q) — the full one-variable model

ARMA(p, q)

$$Y_t = \omega + \underbrace{\phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p}}_{\text{AR: past values}} + \underbrace{\theta_1 e_{t-1} + \cdots + \theta_q e_{t-q}}_{\text{MA: past errors}} + e_t$$

How to read it

- First part is AR: p past values
- Second part is MA: q past errors
- e(t) is today's new shock
- With one variable, ARMA is common

Now we drop the 'one variable' assumption

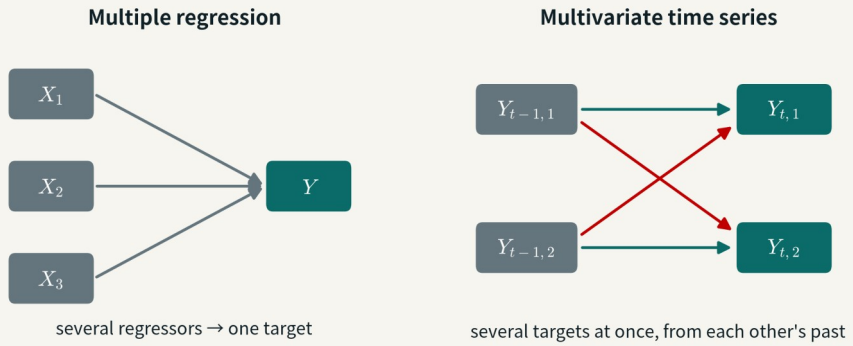
01

VAR model

Vector AutoRegressive — AR for many variables at once

Multivariate ≠ multiple regression

multivariate ≠ multiple — several targets are forecast at the same time



Multiple regression: 1 target

$$\underbrace{Y_t}_{\text{one target}} = \beta_0 + \beta_1 X_{t,1} + \beta_2 X_{t,2} + \beta_3 X_{t,3} + e_t$$

A VAR has many targets, each using every variable's past

A two-variable VAR(1) written out is two regressions

E.g. Y1 = sales, Y2 = ad spend — each equation uses its own and the other's past

VAR(1), two variables — written out

$$Y_{t,1} = \alpha_{0,1} + \alpha_{11,1}\, Y_{t-1,1} + \alpha_{12,1}\, Y_{t-1,2} + e_{t,1}$$
$$Y_{t,2} = \alpha_{0,2} + \alpha_{21,1}\, Y_{t-1,1} + \alpha_{22,1}\, Y_{t-1,2} + e_{t,2}$$

How to read it

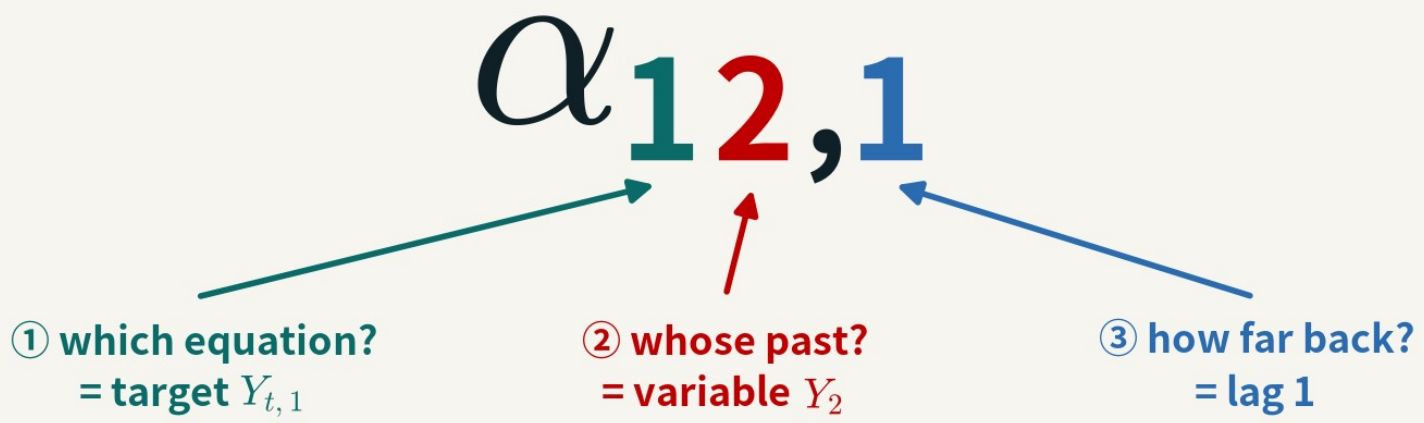
- Top: today's sales from past sales, ads
- Bottom: today's ads from past sales, ads
- α(0,1), α(0,2): intercepts
- e(t,1), e(t,2): equation errors

Each equation is just a regression on both variables' lag 1

The subscripts of α mean 'which equation · whose past · how many lags'

Learn this notation and you can read any size

Note: $\alpha_{0,1}$ = intercept of equation 1



α(12,1) = equation 1's coefficient on variable 2, lag 1

Stack the two equations as matrices: the standard VAR notation

Multiplying out the matrix form gives back the two regressions

Matrix notation — a name for each block

$$\underbrace{\begin{bmatrix} Y_{t,1} \\ Y_{t,2} \end{bmatrix}}_{\mathbf{Y}_t} = \underbrace{\begin{bmatrix} \alpha_{0,1} \\ \alpha_{0,2} \end{bmatrix}}_{\boldsymbol{\alpha}_0} + \underbrace{\begin{bmatrix} \alpha_{11,1} & \alpha_{12,1} \\ \alpha_{21,1} & \alpha_{22,1} \end{bmatrix}}_{A_1} \underbrace{\begin{bmatrix} Y_{t-1,1} \\ Y_{t-1,2} \end{bmatrix}}_{\mathbf{Y}_{t-1}} + \underbrace{\begin{bmatrix} e_{t,1} \\ e_{t,2} \end{bmatrix}}_{\mathbf{e}_t}$$

$$\Longrightarrow \quad \mathbf{Y}_t = \boldsymbol{\alpha}_0 + A_1\,\mathbf{Y}_{t-1} + \mathbf{e}_t$$

How to read it

- Y(t): vector of today's values
- α(0): vector of intercepts
- A1: a table (matrix) of 4 coefficients
- Rows of A1 = equations; columns = whose past
- e(t): vector of errors

A1 is not new — it is the coefficients collected in one table

With k variables, same shape — bigger matrix

k target variables — coefficient matrix A1 is k × k

VAR(1), k variables

$$\begin{bmatrix} Y_{t,1} \\ Y_{t,2} \\ \vdots \\ Y_{t,k} \end{bmatrix} = \begin{bmatrix} \alpha_{0,1} \\ \alpha_{0,2} \\ \vdots \\ \alpha_{0,k} \end{bmatrix} + \begin{bmatrix} \alpha_{11,1} & \cdots & \alpha_{1k,1} \\ \vdots & \ddots & \vdots \\ \alpha_{k1,1} & \cdots & \alpha_{kk,1} \end{bmatrix} \begin{bmatrix} Y_{t-1,1} \\ Y_{t-1,2} \\ \vdots \\ Y_{t-1,k} \end{bmatrix} + \begin{bmatrix} e_{t,1} \\ e_{t,2} \\ \vdots \\ e_{t,k} \end{bmatrix}$$

How to read it

- One row = one equation
- Diagonal: effect of own past
- Off-diagonal: effect of others' past
- In short, the same one line as before

However large, a VAR(1) is one line: Y(t) = α(0) + A1·Y(t−1) + e(t)

Use lags up to p and you get VAR(p)

Just as AR(1) → AR(p), add one coefficient matrix per lag

VAR(1)

$$\mathbf{Y}_t = \boldsymbol{\alpha}_0 + A_1 \mathbf{Y}_{t-1} + \mathbf{e}_t$$

VAR(p): one coefficient matrix Ai per lag

$$\begin{aligned}\mathbf{Y}_t &= \boldsymbol{\alpha}_0 + \sum_{i=1}^{p} A_i \mathbf{Y}_{t-i} + \mathbf{e}_t \\ &= \boldsymbol{\alpha}_0 + A_1\mathbf{Y}_{t-1} + A_2\mathbf{Y}_{t-2} + \cdots + A_p\mathbf{Y}_{t-p} + \mathbf{e}_t\end{aligned}$$

How to read it

- A1: matrix on lag 1
- A2: matrix on lag 2
- Ap: matrix on lag p
- Σ means 'add them all up'

VAR(p) = AR with many variables × many lags

Add an MA part and you get VARMA

ARMA made multivariate — a coefficient matrix B on past error vectors

VARMA(1, 1)

$$\mathbf{Y}_t = \boldsymbol{\alpha}_0 + \underbrace{A_1\,\mathbf{Y}_{t-1}}_{\text{VAR part}} + \mathbf{e}_t + \underbrace{B_1\,\mathbf{e}_{t-1}}_{\text{VMA part}}$$

VARMA(p, q)

$$\mathbf{Y}_t = \boldsymbol{\alpha}_0 + \sum_{i=1}^{p} A_i\,\mathbf{Y}_{t-i} + \mathbf{e}_t + \sum_{j=1}^{q} B_j\,\mathbf{e}_{t-j}$$

How to read it

- A: matrix on past values (VAR part)
- B: matrix on past errors (VMA part)
- Same structure as one-variable ARMA
- The most general form in theory

ARMA : one variable = VARMA : many

VARMA has fewer parameters but is much harder to estimate

VAR (Vector AR) versus VARMA (Vector ARMA)

	VAR	VARMA
Parameters	Many (long lags needed to forecast well)	Few (error terms do much of the work)
Estimation	OLS per equation · simple ML	Iterative ML + derivatives
Difficulty	Easy	Hard — may not converge
In practice	**Mostly VAR for many variables**	**Rare**

Standard: ARMA for one variable, VAR for many

With 5 variables, VAR(12) has 320 parameters, VARMA(1,1) 70

Parameters = intercepts + coefficient matrices + error covariance — VARMA's 70 means 70^2 = 4,900 partial derivatives, repeatedly

Counting parameters

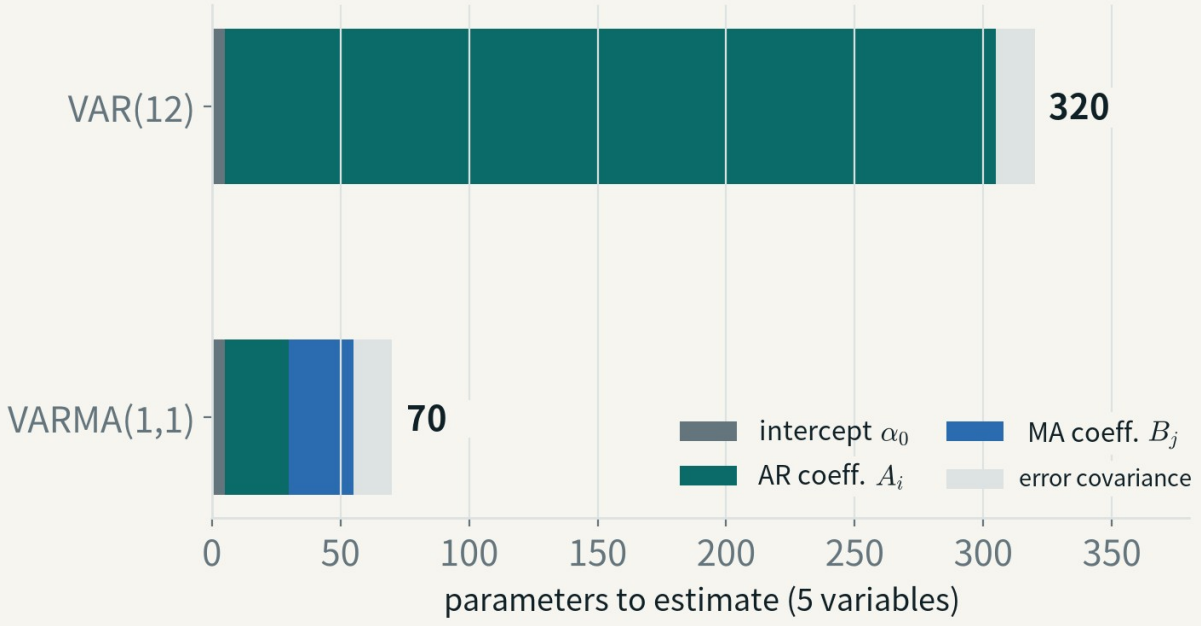
$$\text{VAR}(p): \quad \underbrace{k}_{\alpha_0} + \underbrace{p\,k^2}_{A_1,\ldots,A_p} + \underbrace{k(k+1)/2}_{\text{error cov.}}$$

$$\text{VAR}(12),\ k=5: \quad 5 + 12 \times 25 + 15 = 320$$

$$\text{VARMA}(1,1),\ k=5: \quad 5 + 25 + 25 + 15 = 70$$

How to read it

- VARMA is at least hard to compute



So a VAR uses few variables and lags chosen by information criteria

Pick the lag p with the smallest information criterion

Fit (first term) against simplicity (second) — more lags shrink the first and grow the second

AIC = misfit + complexity penalty

$$\mathrm{AIC}(p) = \underbrace{\ln\left|\hat{\Sigma}_p\right|}_{\text{misfit} \downarrow \text{ as } p\uparrow} + \underbrace{\frac{2\,p\,k^2}{T}}_{\text{penalty} \uparrow \text{ as } p\uparrow}$$

How to read it

- More lags fit better but add complexity
- Pick the p that minimises the sum
- BIC penalises more, so it picks a shorter p
- Software computes it for each p

Set lags by information criteria, not gut

A VAR is estimated by ordinary OLS, equation by equation

Every equation has the same regressors (all variables' past), so separate OLS is efficient

Equation 1 = plain OLS

$$Y_{t,1} = \alpha_{0,1} + \alpha_{11,1}\, Y_{t-1,1} + \alpha_{12,1}\, Y_{t-1,2} + e_{t,1}$$

How to read it

- Run OLS on each of the k equations
- Output: matrices A, error covariance
- Works in Excel or Python
- No iteration needed, unlike VARMA

The biggest strength of a VAR is easy estimation

Before forecasting, check that shocks die out over time

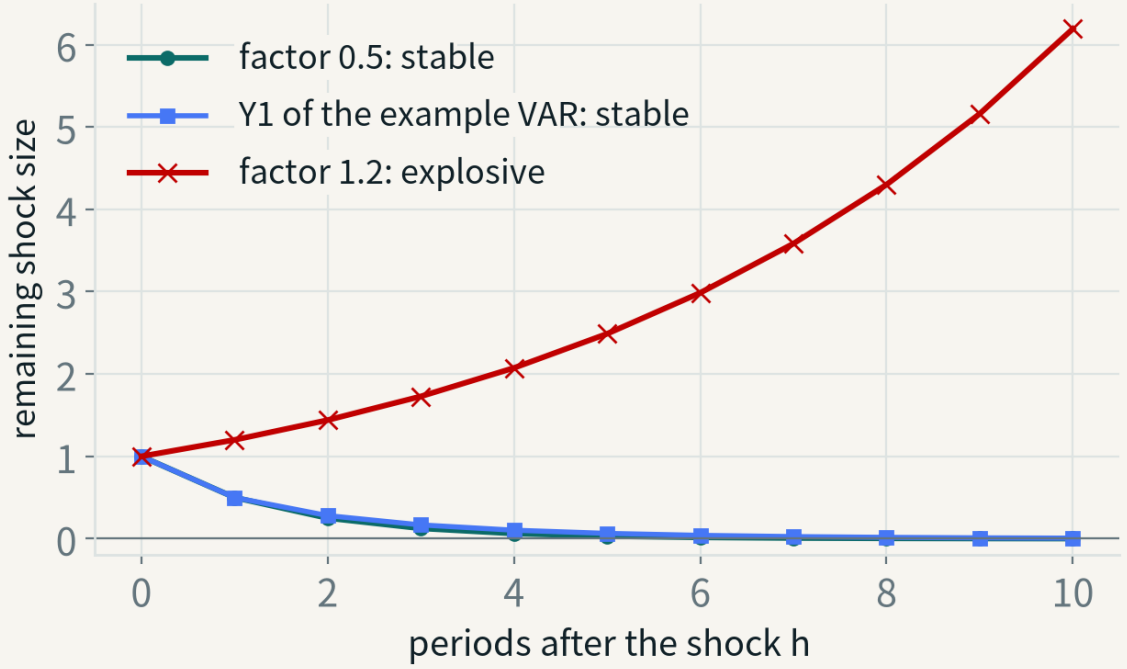
Stability — the factor multiplying a shock each period must be below 1. Required for forecasts and IRFs (Part 3)

Shock growth per period?

$$0.5: \quad 1 \to 0.5 \to 0.25 \to \cdots \to 0$$
$$1.2: \quad 1 \to 1.2 \to 1.44 \to \cdots \to \infty$$

How to read it

- Factor < 1 → goes to 0; for a VAR the factors are A1's eigenvalues



Stability makes forecasts revert to the mean and responses fade to 0

A VAR forecast plugs in yesterday's values and steps forward

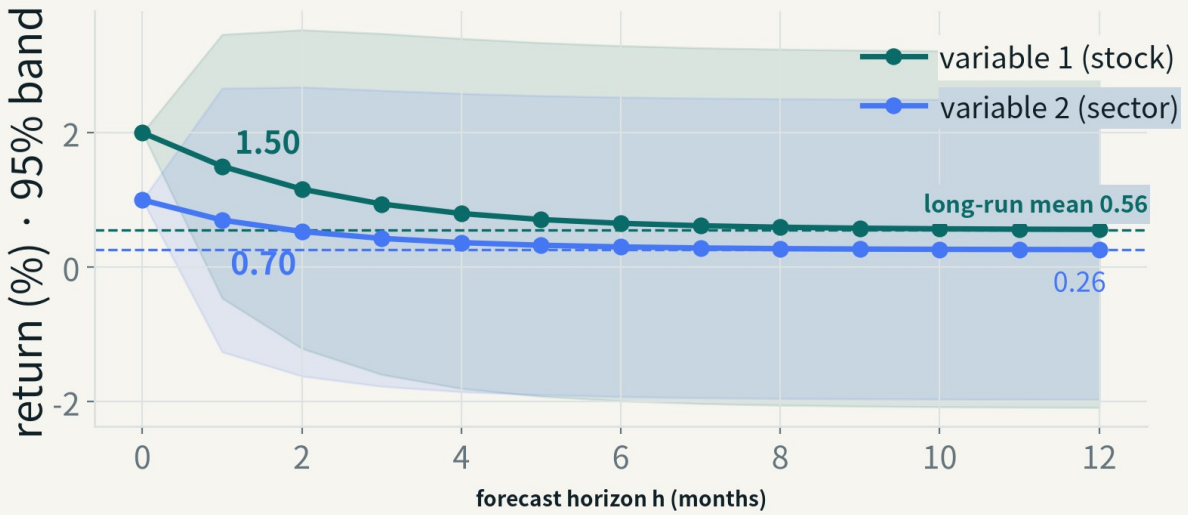
Forecast one step, then feed it back in to forecast two steps

Iterated forecasts

$$\hat{\mathbf{Y}}_{t+1} = \hat{\boldsymbol{\alpha}}_0 + \hat{A}_1 \mathbf{Y}_t$$
$$\hat{\mathbf{Y}}_{t+2} = \hat{\boldsymbol{\alpha}}_0 + \hat{A}_1 \hat{\mathbf{Y}}_{t+1}$$

How to read it

- All variables forecast at once
- Far ahead → long-run mean



A stable VAR's distant forecast is just the mean

02

Granger causality

Does x's past help forecast y?

Granger causality is a simple test: 'does it help forecast?'

Granger causality — not causation in the everyday sense

1 · Not causation

- Inferring cause, unlike everyday cause

2 · Forecast from the past

- If x's past improves y's forecast: causality

3 · Easy to compute

- Two OLS runs — widely used

4 · Don't overread it

- Don't read results as causes

Granger causality = predictive lead, not cause

An ADL model adds another variable's past to its own past

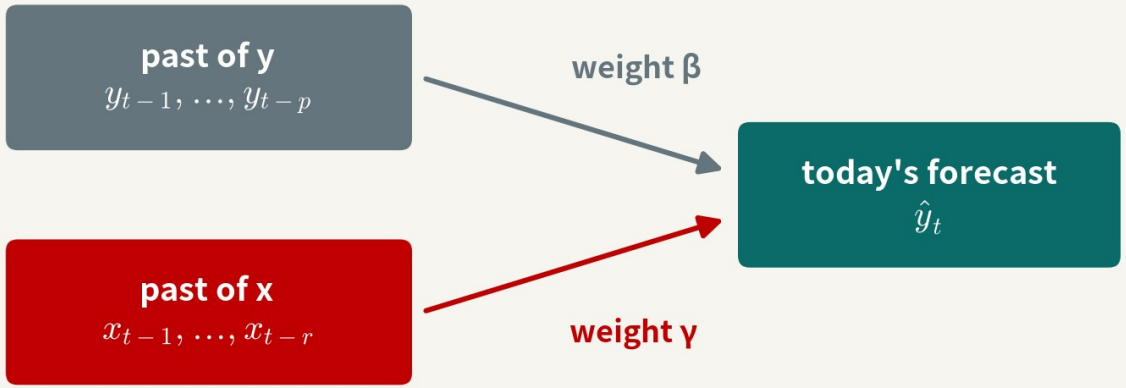
ADL(p, r) = Autoregressive Distributed Lag — includes other regressors besides itself

Forecast = f(both variables' past)

$$\hat{y}_t = F\left(PY_{t-1},\ PX_{t-1}\right), \qquad \varepsilon_t = y_t - \hat{y}_t$$

How to read it

- PY(t−1): all of y's past
- Pick F so the error ε is unrelated to the past



ADL(p, r) is AR(p) plus r lags of x

The usual choice is linear, estimated by plain OLS

Make F a straight line (linear) and it is ordinary regression

Linear forecast equation

$$\hat{y}_t = \alpha + \underbrace{\beta_1 y_{t-1} + \cdots + \beta_p y_{t-p}}_{\text{own past}} + \underbrace{\gamma_1 x_{t-1} + \cdots + \gamma_r x_{t-r}}_{\text{past of } x}$$

ADL(p, r): actual = forecast + error

$$y_t = \hat{y}_t + \varepsilon_t = \alpha + \sum_{j=1}^{p} \beta_j \, y_{t-j} + \sum_{j=1}^{r} \gamma_j \, x_{t-j} + \varepsilon_t$$

How to read it

- β: weights on y's own past
- γ: weights on x's past
- Minimise squared errors → OLS
- p, r: how many lags of each

An ADL is just 'regression with lags'

Fit an ADL for each variable and check the cross coefficients γ

y's equation has x's past (γ); x's equation has y's past (γ*)

Two ADLs: y's equation and x's equation

$$y_t = \alpha + \sum_{j=1}^{p} \beta_j \, y_{t-j} + \sum_{j=1}^{r} \gamma_j \, x_{t-j} + \varepsilon_t$$

$$x_t = \alpha^* + \sum_{j=1}^{p^*} \beta_j^* \, x_{t-j} + \sum_{j=1}^{r^*} \gamma_j^* \, y_{t-j} + \varepsilon_t^*$$

How to read it

- γ ≠ 0 (any one): x helps forecast y
- γ* = 0 (all): y does not help forecast x
- Together: 'x Granger-causes y'
- Each equation is estimated by OLS

x → y needs both γ ≠ 0 and γ* = 0

Stack two ADLs and you get a two-variable VAR

With all lags set to p — one ADL is one row (equation) of the VAR

Two ADLs = VAR(p)

$$\begin{bmatrix} y_t \\ x_t \end{bmatrix} = \begin{bmatrix} \alpha \\ \alpha^* \end{bmatrix} + \begin{bmatrix} \beta_1 & \gamma_1 \\ \gamma_1^* & \beta_1^* \end{bmatrix} \begin{bmatrix} y_{t-1} \\ x_{t-1} \end{bmatrix} + \cdots + \begin{bmatrix} \beta_p & \gamma_p \\ \gamma_p^* & \beta_p^* \end{bmatrix} \begin{bmatrix} y_{t-p} \\ x_{t-p} \end{bmatrix} + \begin{bmatrix} \varepsilon_t \\ \varepsilon_t^* \end{bmatrix}$$

$$\text{row } 1 = \text{ADL of } y, \qquad \text{row } 2 = \text{ADL of } x, \qquad A_j = \begin{bmatrix} \beta_j & \gamma_j \\ \gamma_j^* & \beta_j^* \end{bmatrix} = \begin{bmatrix} \alpha_{11,j} & \alpha_{12,j} \\ \alpha_{21,j} & \alpha_{22,j} \end{bmatrix}$$

How to read it

- β: own past (diagonal)
- γ: other's past (off-diagonal)
- OLS per equation
- Including x(t) is not a VAR

Granger test = are the VAR's off-diagonal coefficients (γ) zero?

An F-test checks the hypothesis that all γ are 0

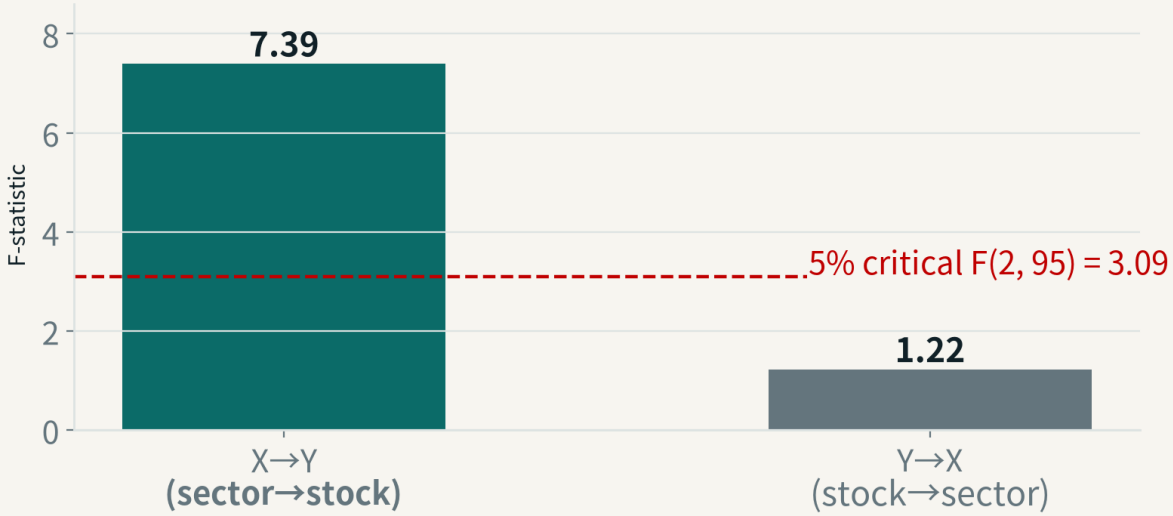
Compare the RSS with and without x's past

Null: x does not Granger-cause y

$$H_0 : \ \gamma_1 = \gamma_2 = \cdots = \gamma_r = 0$$

How to read it

- Large F rejects γ = 0 → it helps
- 7.39 > 3.09 → the sector forecasts the stock
- Reverse: 1.22 → no help



Test both directions to see who leads whom

The F-statistic asks if adding x cuts residuals beyond chance

Numerator: error removed by adding x; denominator: remaining error

F-statistic

$$F = \frac{(\mathrm{RSS}_{\text{without } x} - \mathrm{RSS}_{\text{with } x})/r}{\mathrm{RSS}_{\text{with } x}/(n-p-r-1)}$$

How to read it

- RSS: residual sum of squares (smaller = better)
- r: number of γ tested
- n: number of observations
- F > critical value → reject γ = 0

γ matches the VAR notation — equation 1's coefficient on variable 2, α(12, j)

Testing both directions gives one of four relationships

Combining the tests of x → y and y → x

Four outcomes

- One-way: one side leads
- Feedback: each forecasts other
- Independent: no lagged link

X → Y one-way
X→Y significant · Y→X not



Y → X one-way
Y→X significant · X→Y not



Two-way feedback
Both directions significant



Independent
Neither direction significant



The result is a map of who moves whom first

Case: growth forecast inequality, but inequality barely forecast growth

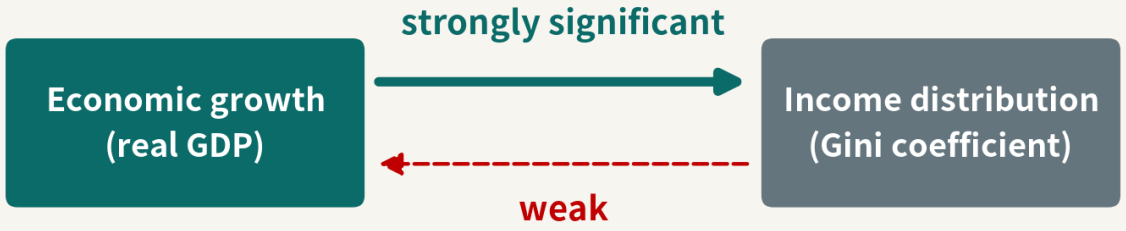
News excerpt — Granger tests on real GDP and the Gini coefficient, annual data 1990–2011

Two ADL results

$$\begin{aligned} \text{Gini}_t \text{ eq.}: &\quad \gamma \text{ (past GDP)} \neq 0 \quad \text{significant} \\ \text{GDP}_t \text{ eq.}: &\quad \gamma^* \text{ (past Gini)} \approx 0 \quad \text{weak} \end{aligned}$$

Read with care

- 22 annual points — a very short sample
- The article's word 'causal' overstates it



Annual data, 1990–2011 (from a newspaper article)

Correct reading: only 'past growth helps forecast inequality'

Granger causality has three limits

Check these before using the test result

LINEARITY

Linearity

- Linear links only
- Misses nonlinear links

STATIONARITY

Stationarity

- Series must be stationary
- Trend? Use differences, growth

HIDDEN CAUSE

Observed variables

- Hidden cause z hits x early, y late
- Then x → y shows up falsely

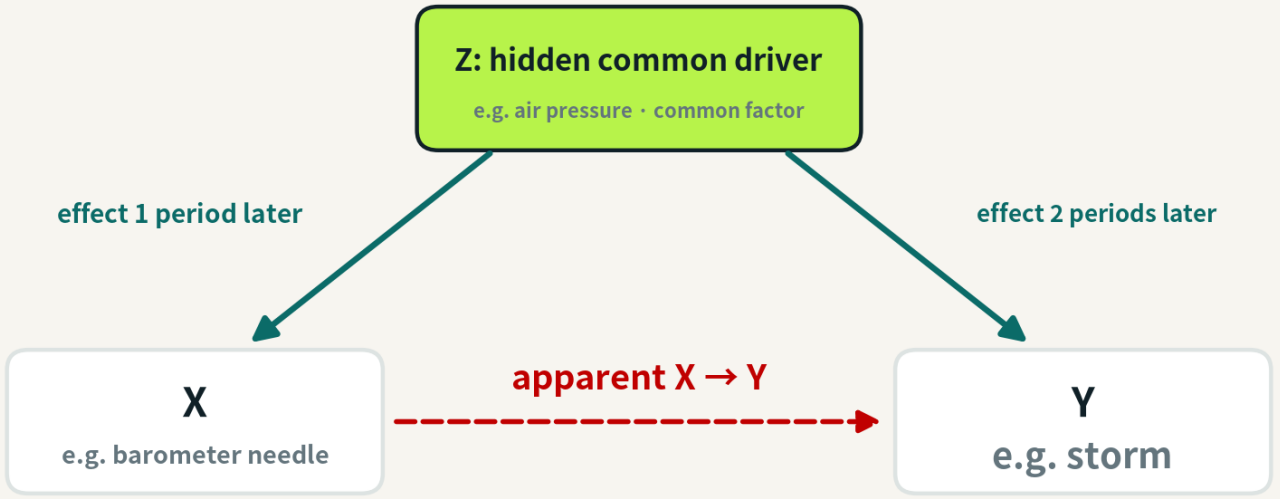
If z is the true cause of x and y, that Granger causality is wrong

Adding the hidden common cause z removes false Granger causality

A barometer (x) 'forecasts' a storm (y); the cause is pressure (z)

How to read it

- z moves x first and y later
- Without z, x → y is significant (F = 1735.8)
- With z, the link vanishes (F ≈ 0)



Omitted variables decide how far to trust it

03

IRF and FEVD

IRF: how a shock spreads · FEVD: who is to blame for forecast error

An impulse response function (IRF) is the path of one shock

IRF = Impulse Response Function — ripples spreading and fading after a stone hits a pond

Definition: with shock minus without shock

$$\begin{aligned}\mathrm{IRF}_{i \leftarrow j}(h) = Y_{t+h,\,i}\ &(\text{with shock } j) \\ - Y_{t+h,\,i}\ &(\text{without shock})\end{aligned}$$

Computing it in a VAR(1) (e: shock vector)

$$\text{response}(h) = A_1 \times \text{response}(h-1) = A_1^h\,\mathbf{e}$$

How to read it

- j: shocked variable; i: responding variable
- h: periods since the shock
- Shock size 1 → response 'per unit'
- Just multiply by A1 once per period

IRF = 'shock one variable by 1: how much does each change h periods later?'

The IRF is a 'function': put in horizon h, get one response

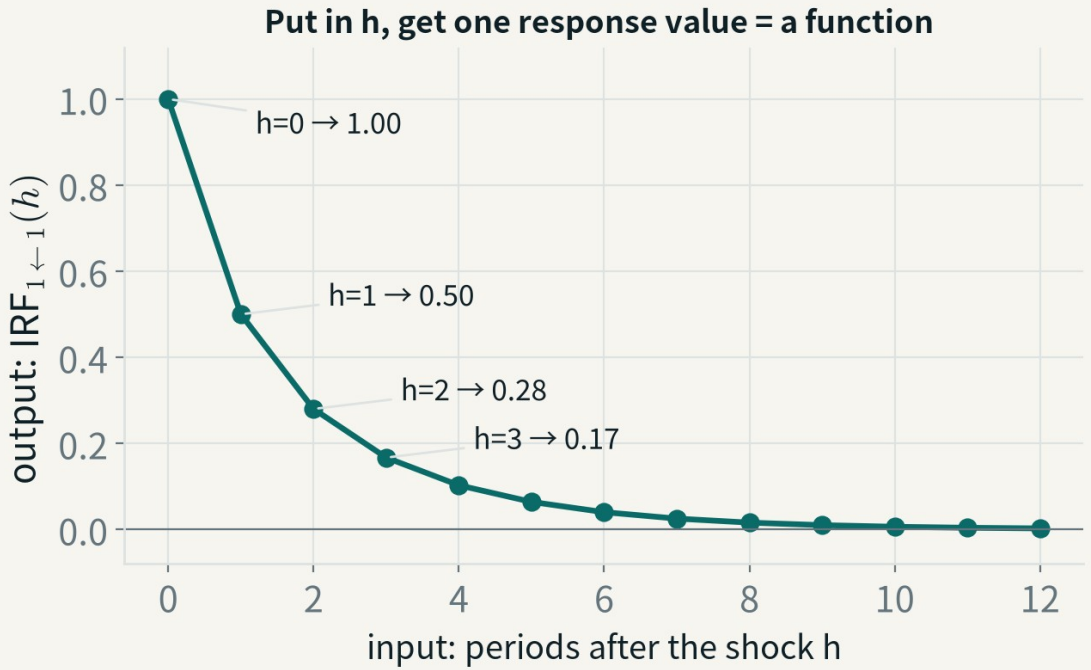
Function = one output per input — input: periods after the shock h; output: the response then

Definition: change in Y(i) h periods after u(j) rises by 1

$$\mathrm{IRF}_{i \leftarrow j}(h) = \frac{\partial Y_{t+h,\,i}}{\partial u_{t,\,j}}, \qquad h = 0, 1, 2, \ldots$$

How to read it

- One value per h → a function of h
- ∂ : 'change per unit change'



The graph of IRF(h) is the 'IRF plot' — hence 'function'

IRF formula: the (i, j) cell of the matrix power is the response

For VAR(p), Ψ(h) is built by multiplying earlier responses by the coefficient matrices

Formula for the IRF

$$\Psi_0 = I, \qquad \Psi_h = A_1\Psi_{h-1} + A_2\Psi_{h-2} + \cdots + A_p\Psi_{h-p}$$
$$\Psi_h = A_1^h \ \ (\mathrm{VAR}(1)), \qquad \Theta_h = \Psi_h P \ \ (\mathrm{orthogonalized})$$
$$\mathrm{IRF}_{i\leftarrow j}(h) = [\Theta_h]_{ij} \ \ (\mathrm{row}\ i,\ \mathrm{column}\ j)$$

How to read it

- Ψ: unorthogonalized
- Θ: orthogonalized
- Row i: responding variable
- Column j: shocked variable

The earlier hand calculation is Ψ(h) = A1 to the power h in a VAR(1)

Multiply by the coefficient matrix step by step to get the IRF

E.g. A1 = [0.5 0.3; 0.1 0.4], shock 1 to Y1 (Y2 is 0) — matrix product = multiply row-wise and add

IRF by hand

$$h = 0: \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$h = 1: \quad \begin{bmatrix} 0.5 & 0.3 \\ 0.1 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.5 \times 1 + 0.3 \times 0 \\ 0.1 \times 1 + 0.4 \times 0 \end{bmatrix} = \begin{bmatrix} 0.50 \\ 0.10 \end{bmatrix}$$

$$h = 2: \quad \begin{bmatrix} 0.5 & 0.3 \\ 0.1 & 0.4 \end{bmatrix} \begin{bmatrix} 0.50 \\ 0.10 \end{bmatrix} = \begin{bmatrix} 0.5 \times 0.50 + 0.3 \times 0.10 \\ 0.1 \times 0.50 + 0.4 \times 0.10 \end{bmatrix} = \begin{bmatrix} 0.28 \\ 0.09 \end{bmatrix}$$

How to read it

- h = 1: Y1 0.50, Y2 0.10
- h = 2: Y1 0.28, Y2 0.09
- h = 3: 0.17, 0.06 …
- Y2 responds via cross coefficient 0.1

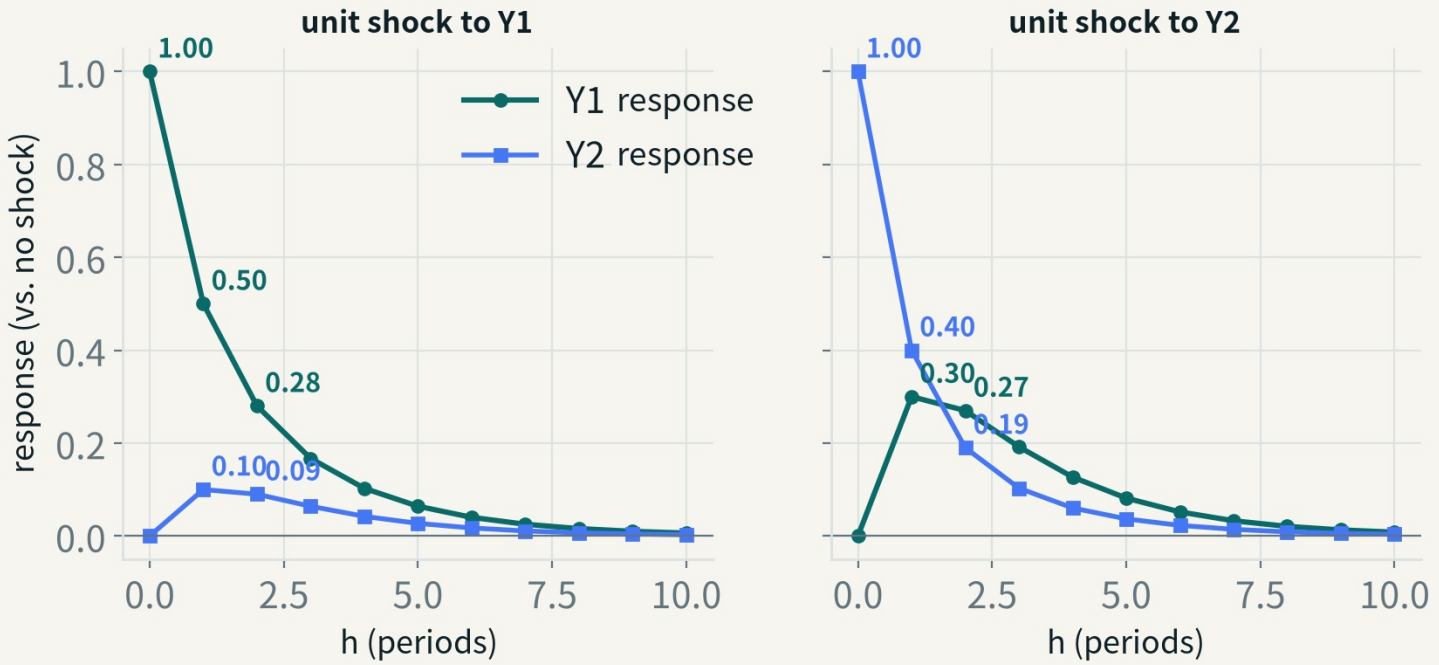
Just keep multiplying the last response by A1

Plot the computed responses to get the IRF chart

Same A1 — left: shock to Y1; right: shock to Y2 (errors assumed unrelated)

How to read it

- h = 0: only the shocked variable is 1
- Others catch it via cross coefficients
- Stable, so all return to 0



An IRF plot answers 'how much, when, how long'

Without orthogonalization you see responses to 'shocks that never happen'

The hand calculation assumed unrelated errors — real stock and sector errors move together (correlation 0.6–0.7)

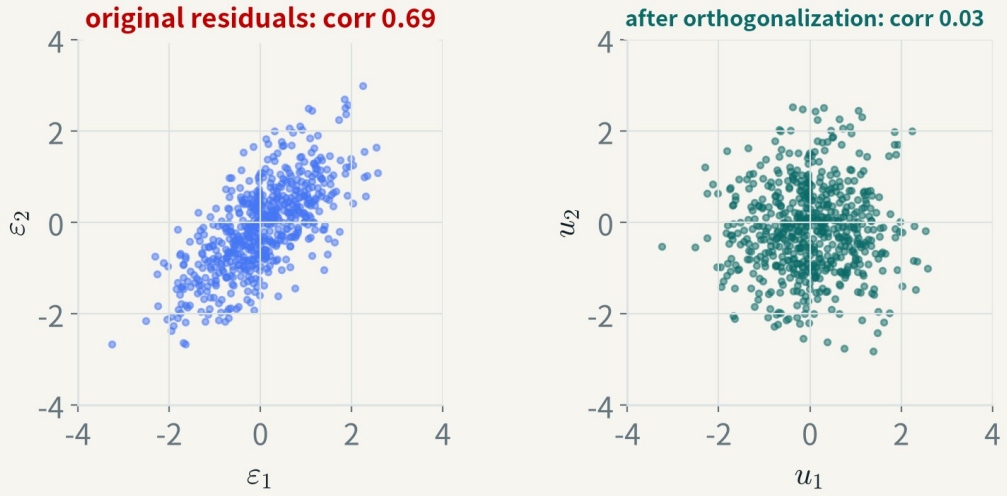
When errors move together

$$\text{Corr}(e_1, e_2) = 0.6 \;\Rightarrow\; \text{when } e_1 = 1, \quad E[e_2] = 0.6$$

$$\text{shock } \mathbf{e} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ assumes } e_2 = 0 \;\Rightarrow\; \text{an unrealistic experiment}$$

Problems if you skip it

- IRF: shocks that never happen
- FEVD: overlap cannot be split



So first assign the overlap to one variable = orthogonalization

Example: split two errors with correlation 0.6, 'sector first'

Orthogonalization (Cholesky) — rewrite as unrelated shocks u1 (sector news), u2 (stock-specific news). Both errors have variance 1

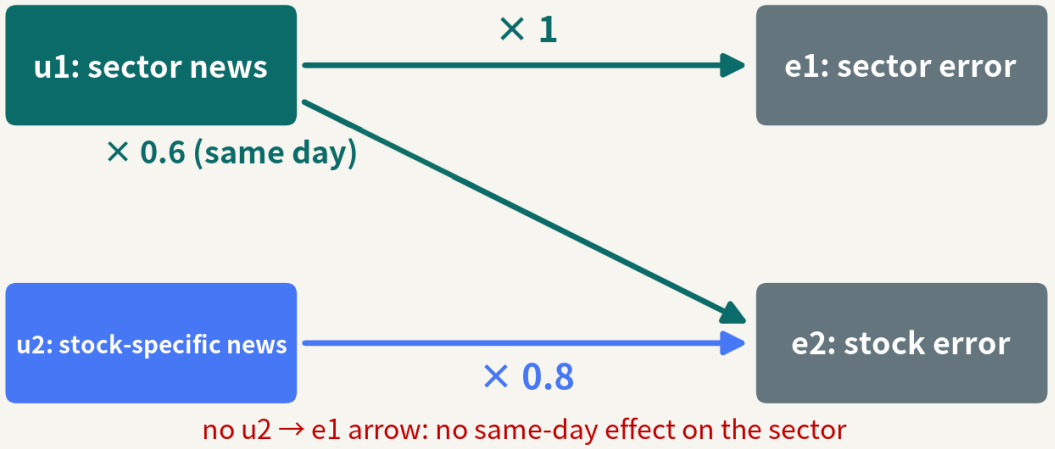
The split and a check

$$e_{t,1} = u_{t,1}$$
$$e_{t,2} = 0.6\,u_{t,1} + 0.8\,u_{t,2}$$

$$\mathrm{Var}(e_2) = 0.6^2 + 0.8^2 = 1$$
$$\mathrm{Cov}(e_1, e_2) = 0.6 \checkmark$$

How to read it

- Sector news moves the stock 0.6 the same day
- Stock news cannot move the sector the same day



The overlap (0.6) goes to the sector, placed first

Recompute the IRF with these shocks

Same A1 = [0.5 0.3; 0.1 0.4] — now 'sector news of 1' moves the stock 0.6 the same day

IRF of orthogonalized shocks (h = 0, 1)

$$u_1 = 1: \quad h{=}0 \begin{bmatrix} 1 \\ 0.6 \end{bmatrix} \xrightarrow{A_1} h{=}1 \begin{bmatrix} 0.5(1) + 0.3(0.6) \\ 0.1(1) + 0.4(0.6) \end{bmatrix} = \begin{bmatrix} 0.68 \\ 0.34 \end{bmatrix}$$

$$u_2 = 1: \quad h{=}0 \begin{bmatrix} 0 \\ 0.8 \end{bmatrix} \xrightarrow{A_1} h{=}1 \begin{bmatrix} 0.5(0) + 0.3(0.8) \\ 0.1(0) + 0.4(0.8) \end{bmatrix} = \begin{bmatrix} 0.24 \\ 0.32 \end{bmatrix}$$

How to read it

- u1: both respond from day one
- u2: sector from next day
- E.g. a day with sector −2, stock +0.5
- Stock error = −1.2 + 0.4 = −0.8

Orthogonalized shocks keep the moves that really happen together

Orthogonalization lets forecast errors split cleanly by shock

Split the stock error's variance of 1 into two shares — with no overlap the shares sum to exactly 100%

Decomposing stock error variance (h = 1)

$$\mathrm{Var}(e_2) = 1 = \underbrace{0.6^2}_{36\% \text{ from } u_1 \text{ (sector)}} + \underbrace{0.8^2}_{64\% \text{ from } u_2 \text{ (own)}}$$

How to read it

- Sector shock share 36%
- Own shock share 64%
- New ordering, new shares

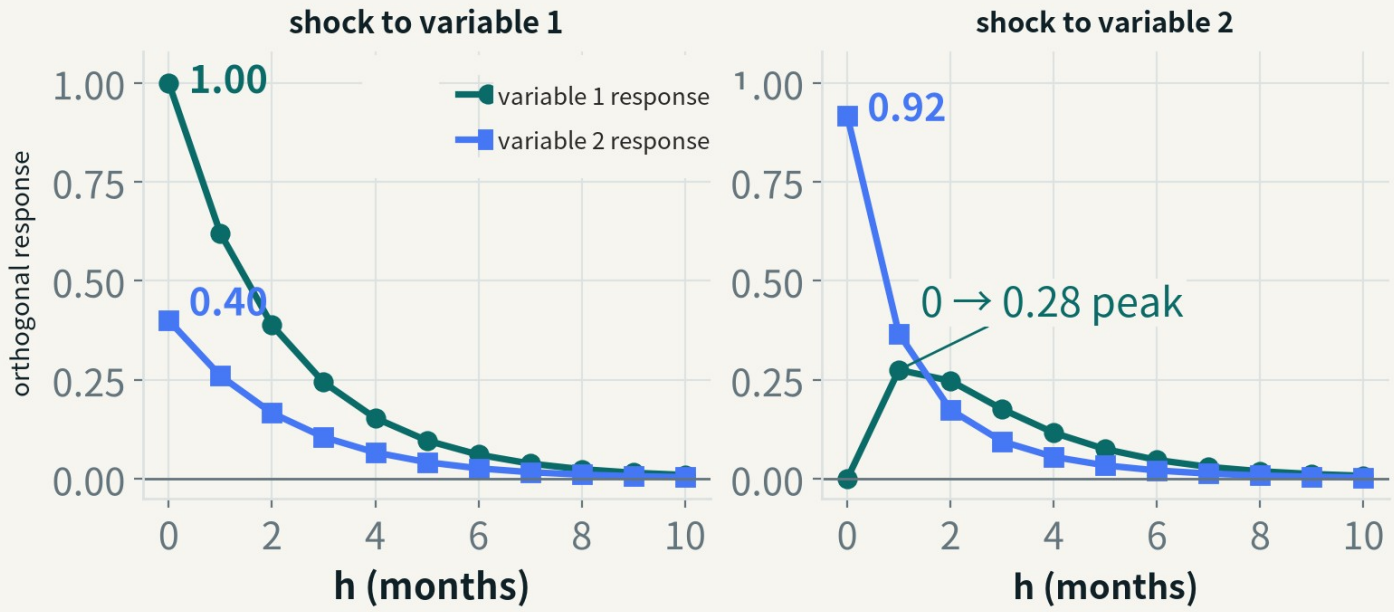
Orthogonalization = sorting out overlap so IRF and FEVD work

Plot with orthogonalized shocks to get a real-analysis IRF

Stock–sector returns — unlike before, same-day response (h = 0) appears

How to read it

- Shock to variable 1: variable 2 responds 0.40 at once
- Shock to variable 2: variable 1 peaks at 0.28 after 1 period
- All return to 0 → stable



Read an IRF the same way — size, timing of peak, duration

Variance decomposition (FEVD) is the share of the h-step forecast error from each shock

FEVD = Forecast Error Variance Decomposition — 'why the forecast missed', split by shock

Forecast error = sum of response (IRF) × shock

$$\underbrace{Y_{t+h,1} - \hat{Y}_{t+h,1}}_{h\text{-step forecast error}} = \sum_{s=0}^{h-1} \left(\underbrace{\psi_{11,s}\, u_1}_{\text{shock 1}} + \underbrace{\psi_{12,s}\, u_2}_{\text{shock 2}} \right)$$

The result: shares summing to 100%

$$\underbrace{100\%}_{\text{forecast error of } Y_1} = \underbrace{90\%}_{\text{own shock}} + \underbrace{10\%}_{\text{shock of } Y_2}$$

How to read it

- ψ: the IRF values found earlier
- u: unrelated shocks (variance 1)
- Why Y1 misses = own shock + other shock
- The IRF is the 'shape'; FEVD is the 'share'

FEVD = 'what % of Y1's forecast error is due to Y2 shocks?'

FEVD is also a 'function': put in horizon h, get a share

Input: horizon h; output: shock j's share of the h-step forecast error — a ratio of summed squared IRFs

Formula: ratio of squared IRF sums

$$\mathrm{FEVD}_{i\leftarrow j}(h) = \frac{\sum_{s=0}^{h-1} \theta_{ij,s}^2}{\sum_{s=0}^{h-1} \sum_{k=1}^{K} \theta_{ik,s}^2}, \qquad \theta_{ij,s} = \mathrm{IRF}_{i\leftarrow j}(s)$$

Properties: between 0 and 1, always sums to 1

$$0 \le \mathrm{FEVD}_{i\leftarrow j}(h) \le 1, \qquad \sum_{j=1}^{K} \mathrm{FEVD}_{i\leftarrow j}(h) = 1 \ \text{ for every } h$$

How to read it

- θ: orthogonalized IRF values
- Numerator: error from shock j
- Denominator: error from all shocks
- Shares change as h changes

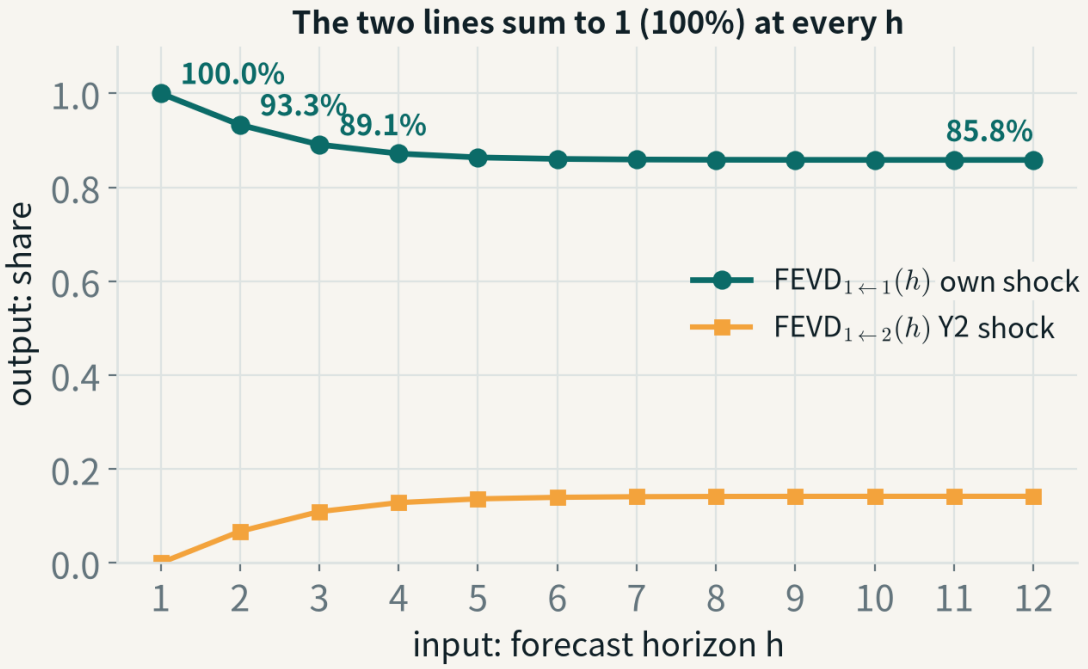
FEVD(h) is a second function built from IRF(h)

Plot the FEVD function to see shares change with horizon

Same A1 — Y1's forecast error, horizons h = 1–12

How to read it

- h = 1 → 100%
- h = 2 → 93.3%
- h = 3 → 89.1%
- h = 10 → 85.8%
- The two lines always sum to 100%



In a stable VAR, FEVD converges as h grows

FEVD: square the IRF values, sum, then take the ratio

Same A1, Y1's forecast error — uses own response ψ11 and cross response ψ12

FEVD by hand (Y1's own-shock share)

$$\psi_{11}: 1,\ 0.50,\ 0.28 \qquad \psi_{12}: 0,\ 0.30,\ 0.27 \quad (\text{IRF of } Y_1)$$

$$h=1: \quad \frac{1^2}{1^2+0^2} = 100\%$$

$$h=2: \quad \frac{1^2+0.5^2}{(1^2+0.5^2)+(0^2+0.3^2)} = \frac{1.25}{1.34} = 93.3\%$$

$$h=3: \quad \frac{1.25+0.28^2}{1.34+0.28^2+0.27^2} = \frac{1.328}{1.491} = 89.1\%$$

How to read it

- h = 1: 100%
- h = 2: 93.3%
- h = 3: 89.1%
- The rest is the Y2 shock share

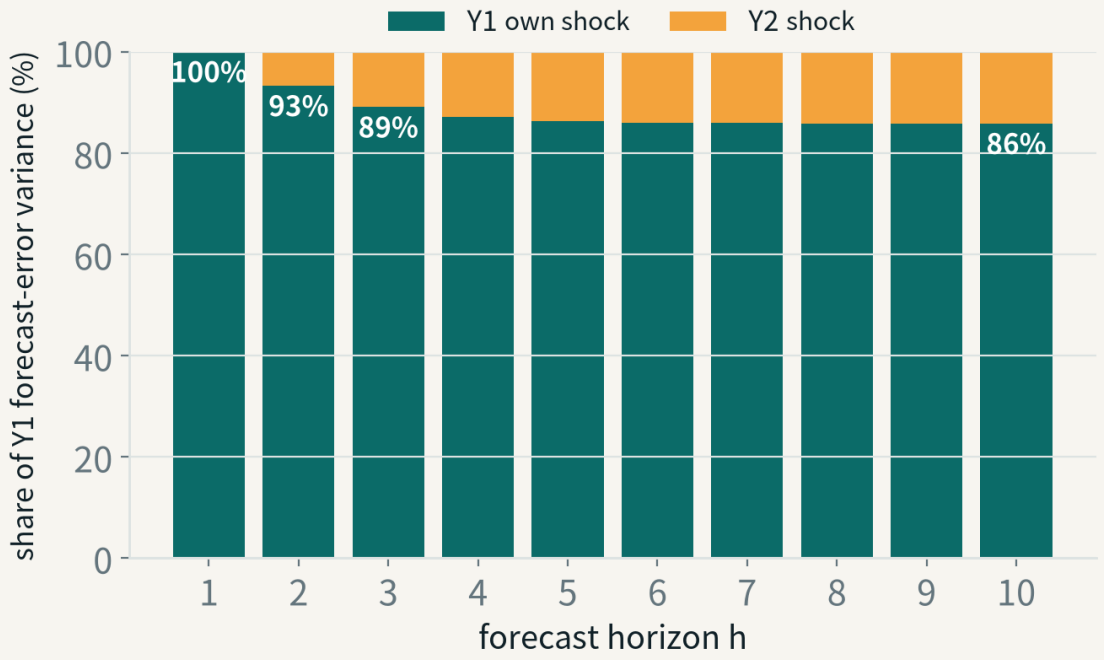
Longer horizons give others' shocks more share

Stack shares over horizons to get the FEVD chart

Same A1 — sources of Y1's forecast error, horizons 1–10

How to read it

- h = 1: own shock 100%
- h = 10: own 86%, Y2 14%
- Always sums to 100%



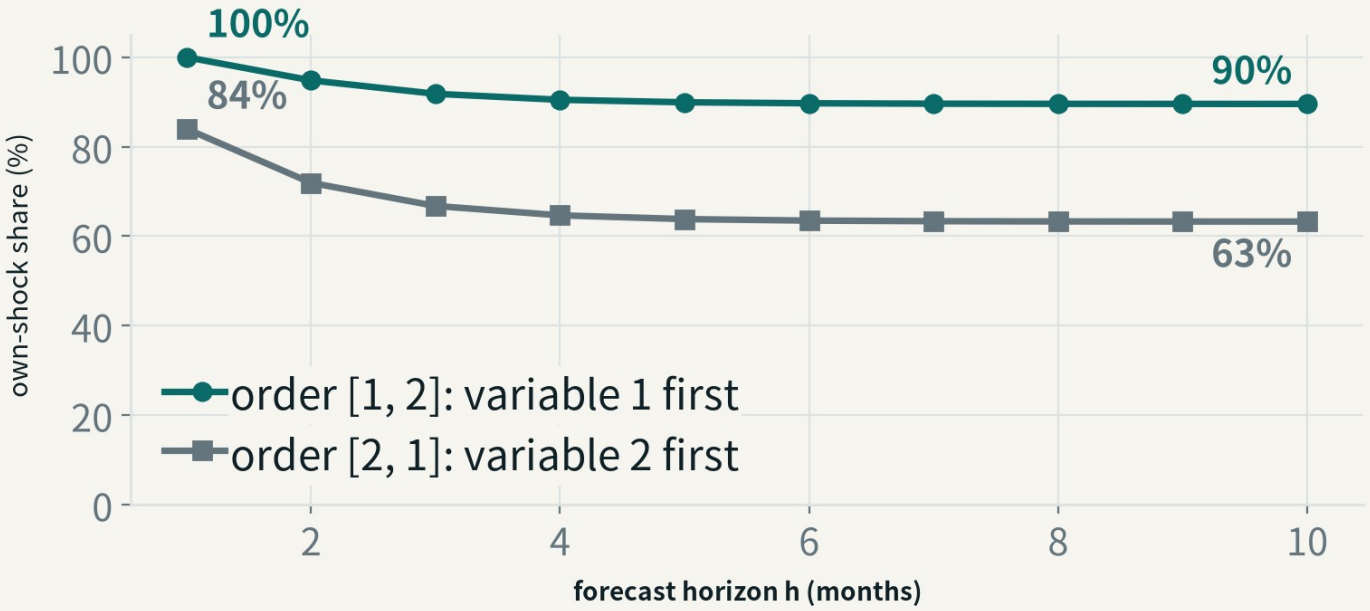
Short horizons: own shock; longer horizons mix in others

But changing only the ordering moves the shares a lot

Stock–sector example, same data, Cholesky order [1, 2] vs [2, 1]

How to read it

- Order [1, 2]: own share 90%
- Order [2, 1]: own share 63%
- Always state the ordering with results



Report IRF · FEVD with their variable ordering

04

Synthesis

One VAR, three questions

One VAR covers forecasts · lead–lag · shock paths

One model; the tool depends on the question

Tool	Question	What to read
VAR forecast	**What happens next?**	**All variables jointly**
Granger causality	**Does another series' past help?**	**F-test of cross coefficients = 0**
IRF	**How does a shock spread?**	**Response path over time**
FEVD	**Who drives forecast error?**	**Share by shock source (sums to 100%)**

Next week: level variables that drift together — cointegration

AR · MA — one variable

VAR — many variables in one line

Granger · IRF · FEVD — questions for a VAR

Next week
Cointegration, ECM